Analyzing data on the level of cognitive processes - ESCoP 2025

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Exercise 7: Signal Discrimination Model

In this exercise, we will have a look at fitting the Signal Discrimation model to a continous reproduction task varying the set size of to be remembered color patches.

The data is part of the `bmm` package and can be loaded as `oberauer_lin_2017`.

Your task is:

1. to specify the `sdm` model object
2. estimate an `sdm` model evaluating if the parameter `c` or `kappa` changes over set size. Consider entering set size as a numeric variable assuming a linear set size effect
3. fit the `sdm` to the data
4. plot the predicted effects on `c` and `kappa` using the `conditional_effects` function
5. use the `hypothesis` function to test for effects of set size on `c` and `kappa`

If you want compare your results to fitting a two-parameter mixture model on the data. Again, see how the proportion of responses from memory and kappa is affected by the set size variable.